Saumya Chaudhary

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EDUCATION

Master of Science in **Computer Science** | *University of Illinois, Chicago, IL* **August 2024 – May 2026**
Coursework: Artificial Intelligence-I, Introduction to Data Science, Big Data Mining, Responsible AI, Artificial Intelligence II GPA: 4.0

Bachelor of Technology in **Computer Science** | *Medicaps University, Indore, India* **August 2020 – May 2024**
Coursework: Software Engineering, Computer Networks, Data Structure, Object-Oriented Programming GPA: 3.7

TECHNICAL SKILLS

Programming Languages: Python, C#, Java, C++, JavaScript, R
Libraries & Frameworks: NumPy, Pandas, Scikit-learn, Matplotlib, TensorFlow, Keras, OpenCV, PyTorch, ASP.NET, Flask, Seaborn, GANs, .NET, Apache Spark, NGINX, Apache Kafka, Azure AI, Chart.js
Machine Learning: CNN, Linear Regression, Logistic Regression, Decision Trees, Random Forest, XGBoost, SVM, Naive Bayes, K-Means
Natural Language Processing (NLP): LangChain, Prompt Engineering, HuggingFace, LoRA, RAG
Tools & Platforms: Docker, ElasticSearch, Git, Gitlab, GitHub, Postman, Linux, TensorBoard, Power BI, GCP, CUDA, MLflow
Database: SQL, MySQL, Microsoft SQL Server, MongoDB, PostgreSQL

WORK EXPERIENCE

Computer Specialist at **School of Public Health, UIC |** *Chicago, IL* **April 2025 – Present**

- Developed an interactive research dashboard using C#, Chart.js, and SQL Server to visualize inter-author collaboration networks, enabling 500+ stakeholders to evaluate research influence and strategic partnerships.
- Engineered a competency management portal with C#, SQL Server, and JavaScript to track versions, manage approvals, and perform CRUD operations—automating workflows and reducing manual updates by 90%.

Software Engineering Intern at **Swastika Investmart Ltd |** *Indore, India* **January 2024 – July 2024**

- Created a scalable backend for a stock information dashboard using Flask and MongoDB to serve data for over 4,500 stocks; implemented Docker-based cron jobs for scheduled tasks and configured NGINX to secure and optimize API response delivery.
- Optimized 20+ email templates and C# APIs integrated in customer acquisition workflows by resolving formatting and delivery issues, leading to a smoother onboarding process and a 15% reduction in customer setup time.
- Designed and deployed C# APIs to automate the generation of daily trade reports for 150 stocks, streamlining the research team's workflow and accelerating trade call evaluations by 30%.

Research Intern at **Space Application Center, ISRO |** *Ahmedabad, India* **June 2023 - August 2023**

- Reconstructed the missing blue band in ~1 million satellite images using red, green, and near-infrared channels, addressing a critical multispectral image completion challenge.
- Designed and trained a custom GAN architecture from scratch (TensorFlow, Keras), achieving 94% accuracy over 500 epochs—outperforming legacy methods for blue band prediction.
- Integrated Dense Blocks into a U-Net backbone, surpassing the baseline pix2pix model with a 97% R^2 score and 0.013 FID; validated improvements through quantitative metrics and visual assessments.

PROJECTS

StudyMate: Agentic Tutor

- Configured an agentic RAG system using LangChain and LLaMA-3B to autonomously route user queries between summarization, document-based Q&A, and generic interactions; supported ingestion of multiple documents across PDF, DOCX, and PPT formats.
- Implemented FAISS with HuggingFace embeddings and a recursive chunking pipeline to support fast semantic search on documents up to 500 pages; integrated auto-generated MCQs and short-answer quizzes to evaluate comprehension.

Music Recommendation System

- Produced a lyric-based recommendation engine to address the cold-start problem using 10,000+ songs from Last.fm and Million Song Dataset, focusing on semantic similarity via lyrics.
- Engineered song embeddings using Word2Vec, graph-based learning, and Sentence-BERT; used XGBoost for artist-tag prediction, improving cold-start recommendation accuracy by 54%.

Wildfire Prediction Model

- Built ML classifiers to predict wildfire causes and size categories using 300,000+ demographic and environmental records spanning 30 years
- Achieved 70% accuracy with Random Forest and improved to 73% with SMOTE to address class imbalance.